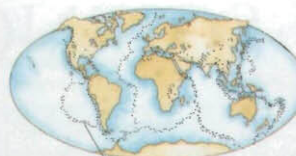


EARTHQUAKES



FROM A GENTLE RIPPLE to terrifying and violent movements in the Earth, earthquakes literally rock the world. Earthquakes are tremors in the ground, created by the sudden movement of tectonic plates – huge slabs of rock that make up the Earth's crust. The majority of earthquakes are so gentle that no one notices them, but some are so violent they destroy whole cities. An earthquake's effect and intensity are measured on different scales. In earthquake-prone countries, planning minimizes the damage earthquakes cause.



Earthquake zones

Earthquake zones
Although earthquakes can occur anywhere, they are more frequent in earthquake zones. These zones, such as Japan and California, lie near the moving margins of the tectonic plates, called fault lines.

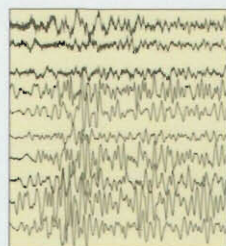
What is an earthquake?

Tectonic plates usually slide past each other, but sometimes they get stuck together. The stress on the rocks builds up until they fault (crack). The tectonic plates then jolt past each other, sending shock waves through the ground. These vibrations, known as seismic waves, cause the earth to quake.



Epicentre

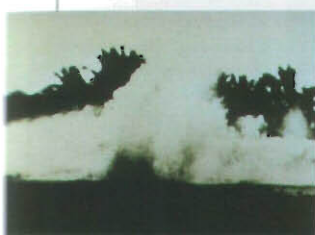
The point at which an earthquake occurs is known as the focus. Above the focus is the epicentre – the point on the Earth's surface where the effects of an earthquake are most devastating. The focus may be as much as 700 km (185 miles) below the epicentre. In 1985, an earthquake in Mexico City, with its epicentre in the Pacific Ocean, left 9,500 people dead. It measured 8.1 on the Richter scale.



Seismometer

Seismometers show seismic waves, and measure an earthquake's location and intensity on the Richter scale. The height of each line shows the wave's force.

Reading from Kobe, Japan



Tsunami

These are huge waves precipitated when an earthquake or volcanic eruption shakes the sea floor. Tsunamis roll along the ocean floor as fast as a jet plane. When they reach shallow coastal waters, they rear up into water ridges about 30 m (100 ft) high. Many tsunamis occur in the Pacific Ocean, such as the one in Hawaii, 1964 (left).

Earthquake proofing

Technology cannot prevent earthquakes but it can help limit their damage, particularly in building design. Most loss of life is caused not by the shaking ground, but by the collapse of buildings and roads, and fires started by damage to electrical equipment.



Building design

Pyramid-shaped, curved, and fire-resistant buildings and structures, such as this staircase in California, USA, bend rather than break during an earthquake. Mounting foundations on rubber also helps absorb some of the earthquake shocks.

Timeline

1556 Reports of an earthquake in the region of Shaanxi, China. Almost a million deaths.

1755 Lisbon, Portugal, is destroyed by an earthquake and the subsequent flood.

1883 Krakatoa Island destroyed by earthquake and tsunami.

1906 Quake flattens San Francisco, USA.

1964 Alaska hit by a very severe earthquake, measuring 9.2 on the Richter scale.

1964 Earthquake in Alaska generates a tsunami, which causes damage as far away as California, USA.

1976 Earthquake in China kills 255,000.

1990 In Iran 40,000 people die in quake.

1995 Kobe, central Japan, is devastated by an earthquake.

1999 Turkish quake kills 20,000 and makes 200,000 homeless.

2001 Earthquake in Gujarat, India, leaves 30,000 people dead.

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